



4.1 THE PURPOSE OF THE PHYSICAL LAYER

4.1.1 THE PHYSICAL CONNECTION

A physical connection is required for network communications to occur.

- In a wired network, the connection is made using a cable.
- In a **wireless** network, the connection is made using radio waves.

The type of connection depends on the network:



- Many corporate offices and enterprises have desktop or laptop computers that are physically connected with cables to a shared switch.



- Businesses also use wireless connections for laptops, tablets, and smartphones.

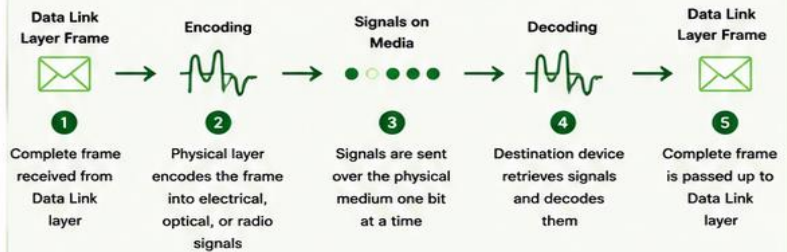


Devices on a wireless network must be connected to a wireless access point (AP) or wireless router.

4.1.2 THE PHYSICAL LAYER

- Provides the means to transport the bits of a data link layer frame across the network media.
- Accepts a complete frame from the data link layer and encodes it as a **series** of signals that are transmitted to the **local media**.
- Encoded bits are received by an end device or an intermediate device.
- The destination device retrieves the signals from the media, converts the bits back to their original form, and passes the complete frame up to the data link layer.

ENCAPSULATION AT THE PHYSICAL LAYER



The physical layer creates the electrical, optical, or radio wave signals that represent the bits in each frame.

PHYSICAL LAYER HARDWARE AND CONNECTIONS

WIRELESS ROUTER



- 1 Antenna(s) (wireless models only) – Used to send and receive radio signals.
- 2 Ethernet (LAN) Ports – Connect devices on the local area network (LAN) using Ethernet cables.
- 3 Internet (WAN) Port – Connects the router to the Internet or another network device (such as a modem).

WIRED CONNECTION TO WIRELESS ROUTER



Network Interface Cards (NICs)

NICs connect a device to the network.

Ethernet NICs are used for wired connections. Wireless NICs are used for wireless connections.

Most devices have one built-in NIC. A device may also have additional NICs such as Ethernet, wireless, cellular (4G/5G), Bluetooth, or other options.



KEY POINT

Similar to corporate offices, most homes offer both wired and wireless connectivity to the network. This means you can have a home router and a laptop connecting to the local area network (LAN).

WIRED CONNECTION USING AN ETHERNET NIC



An Ethernet NIC provides a wired connection between a device and the network using an Ethernet cable. It is common in desktops, laptops, and many other devices such as tablets and smartphones.



TAKEAWAY

The physical layer is all about the physical connection and the signals that carry the data across the medium.

4.2 PHYSICAL LAYER CHARACTERISTICS (PART 1)

4.2.1 PHYSICAL LAYER STANDARDS

The physical layer deals with the electronic circuitry, media, connectors, and signaling that carry bits (1s and 0s) across the network.



Protocols and operations of the upper OSI layers are performed in software.



Services and protocols in the TCP/IP suite are defined by the Internet Engineering Task Force (IETF).



The physical layer standards define the hardware components and how electrical and communication signals are transmitted.



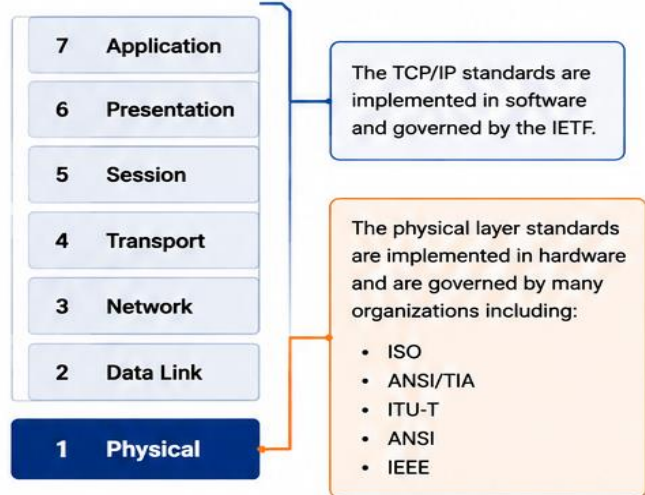
INTERNATIONAL STANDARDS ORGANIZATIONS

- **ISO** – International Organization for Standardization
- **TIA/EIA** – Telecommunications Industry Association / Electronic Industries Association
- **ITU** – International Telecommunication Union
- **ANSI** – American National Standards Institute
- **IEEE** – Institute of Electrical and Electronics Engineers
- **FCC** – U.S. Federal Communication Commission
- **ETSI** – European Telecommunications Standards Institute



Other regional bodies, such as CSA (Canadian Standards Association), CENELEC (European Committee for Electrotechnical Standardization), and JSA/JIS (Japanese Standards Association), develop local specifications.

PHYSICAL LAYER & THE OSI MODEL



4.2.2 PHYSICAL COMPONENTS

1. CIRCUITRY



Electronic components on network devices that generate, transmit, and process signals.

2. MEDIA



The physical path over which signals travel (copper cable, fiber-optic cable, or wireless).

3. CONNECTORS



Interfaces that connect cables and devices together.

4. SIGNALING



The method used to represent bits as electrical, optical, or radio signals (1s and 0s).

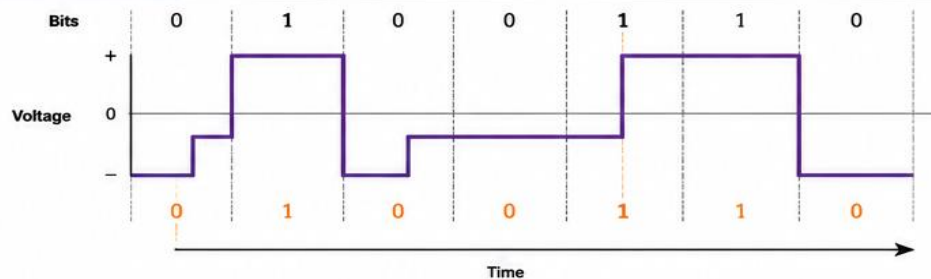
4.2.3 ENCODING

Encoding is the process of converting a series of bits (1s and 0s) into signals suitable for transmission on the media.

Different encoding schemes change the signal level to represent binary data.



The figure shows an example of line coding where the voltage level represents binary 1s and 0s.



4.2.4 SIGNALING

Signaling is the method used by the physical layer to generate signals that represent binary 1s and 0s on the media.

- A long pulse or higher level might represent a 1.
- A short pulse or lower level might represent a 0.

It is similar to Morse code, which uses on/off tones or lights to send information.



Click each tab to see how 1s and 0s are represented on different media.

Copper Cable

Fiber Optic Cable

Wireless Media

Electrical Signals Over Copper Cable



Bit	0	1	0	1	0
Signal					
Meaning	No pulse (Low)	Pulse (High)	No pulse (Low)	Pulse (High)	No pulse (Low)



KEY TAKEAWAY

The physical layer provides the physical connection, defines how bits are encoded and signaled, and uses standards to ensure devices can communicate across different media.

4.2 PHYSICAL LAYER CHARACTERISTICS (PART 2)

– Bandwidth & Bandwidth Terminology

4.2.5 BANDWIDTH

What is Bandwidth?



Bandwidth is the **capacity** of a medium to carry data (bits) from one place to another in a given time.



Key Points

- Digital bandwidth measures how much data can flow per unit time.
- Units: kbps, Mbps, Gbps
- Bandwidth is the number of bits transmitted per second.
- Sometimes called the **speed** that bits travel (not always accurate).
- Example: In 10 Mbps Ethernet, bits are sent at the speed of electricity.



Factors Affecting Practical Bandwidth

- Properties of the physical media
- Technologies used for signaling and detecting network signals
- Physical media, current technology and laws of physics all affect the available bandwidth.

More Bandwidth → Faster Data Transfer



Common Units of Bandwidth

Unit of Bandwidth	Abbreviation	Equivalence
Bits per second	bps	1 bps = fundamental unit of bandwidth
Kilobits per second	Kbps	1 Kbps = 1,000 bps = 10^3 bps
Megabits per second	Mbps	1 Mbps = 1,000,000 bps = 10^6 bps
Gigabits per second	Gbps	1 Gbps = 1,000,000,000 bps = 10^9 bps
Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^{12} bps

4.2.6 BANDWIDTH TERMINOLOGY



Terms used to measure the quality of bandwidth: **Latency** | **Throughput** | **Goodput**



1. Latency

- Time (including delays) for data to travel from one point to another.
- In networks with multiple segments, throughput cannot be faster than the slowest link in the path.
- Even if most links have high bandwidth, one low-throughput segment creates a bottleneck for the whole network.



2. Throughput

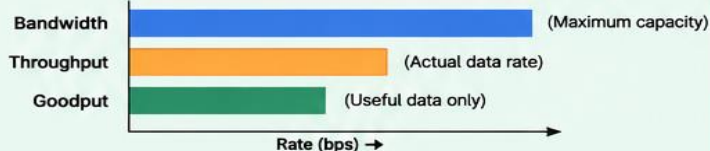
- Actual measure of bits transferred over the media in a given period of time.
- Usually lower than the specified bandwidth.
- Many factors affect throughput:
 - Amount of traffic
 - Type of traffic
 - Latency caused by devices between source and destination
- Online speed tests show throughput.



3. Goodput

- Measures **useful data** actually transferred per unit time.
- Goodput = Throughput – traffic overhead (sessions, acknowledgments, encapsulation, retransmitted bits, etc.)
- Always lower than throughput, and generally lower than bandwidth.

Bandwidth vs Throughput vs Goodput



Example – Online Speed Test



Quick Takeaway

Bandwidth is the **maximum capacity** of the media, while latency, throughput and goodput are used to assess the real performance and quality of a network connection.



4.3 COPPER CABLING (PART 1)

4.3.1 Characteristics of Copper Cabling



- Copper cabling is the most common type of cabling used in networks.
- It supports both voice (telephone) and data (computer) communication.
- Copper media is cheaper than optical cabling.
- Data is carried as electrical signals.

4.3.2 Types of Copper Cabling

UTP Cable (Unshielded Twisted-Pair)



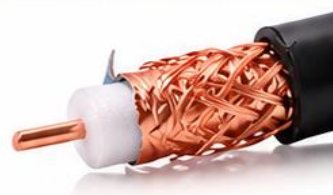
- No shielding around the wires.
- More flexible and cheaper.
- Common in LANs (Ethernet).

STP Cable (Shielded Twisted-Pair)



- Has an overall shield (foil or braided) to reduce interference.
- Used where EMI is a problem.

Coaxial Cable



- Single central conductor.
- Shielded to reduce noise.
- Used for cable TV and internet.

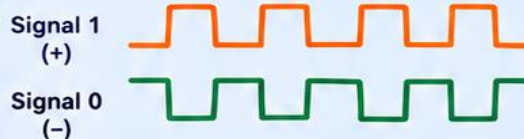


Copper Cable Terminology

Term	Meaning
UTP	Unshielded Twisted-Pair cable.
STP	Shielded Twisted-Pair cable.
Coaxial Cable	Cable with a central conductor and shielding.



Twisted-Pair Cable Signals



- Twisting the pair of wires helps to:
- ✓ Reduce interference (noise)
 - ✓ Improve signal quality



KEY TAKEAWAY

Copper cabling (UTP, STP and coaxial) is a low-cost, widely used medium for network communication. UTP is flexible and cheaper, STP provides shielding, and coaxial cable is used for TV and internet.



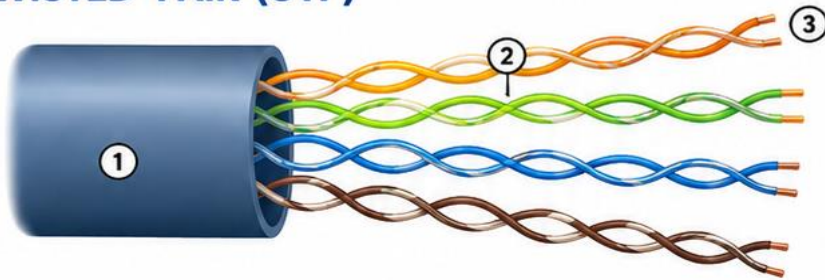
4.3 COPPER CABLING (PART 2)

4.3.3 UNSHIELDED TWISTED-PAIR (UTP)

- UTP is the most common networking media.
- Uses RJ-45 connectors for network devices (switches, routers, hosts).
- In LANs, UTP has four pairs of color-coded wires twisted together and covered by a flexible plastic sheath.
- Twisting reduces signal interference between wires.

KEY POINT

Twisting + color codes help identify each wire and pair and aid in proper cable termination.



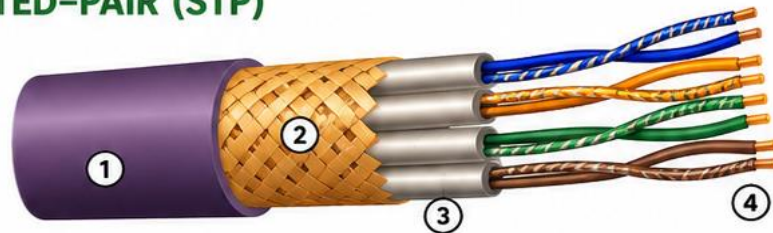
- 1 **Outer Jacket** – protects the copper wires from physical damage.
- 2 **Twisted Pairs** – reduce interference (EMI and crosstalk) from other wires.
- 3 **Color-coded Plastic Insulation** – electrically isolates wires from each other and identifies each pair.

TYPICAL COLOR CODE (T568B)

Pair 1	White/Orange – Orange
Pair 2	White/Green – Green
Pair 3	White/Blue – Blue
Pair 4	White/Brown – Brown

4.3.4 SHIELDED TWISTED-PAIR (STP)

- STP provides better noise protection than UTP.
- More expensive and harder to install.
- Uses RJ-45 connectors.
- Combines shielding to reduce EMI and RFI and to counter crosstalk.
- Shielded data connectors are used for maximum protection.
- If not grounded properly, the shield can act like an antenna and pick up unwanted signals.



- 1 **Outer Jacket**
- 2 **Braided or Foil Shield** – overall shield reduces EMI/RFI and crosstalk.
- 3 **Foil Shields** – each pair wrapped in foil for additional protection.
- 4 **Twisted Pairs** – four pairs of wires.



TIP

Proper grounding of the shield is essential. Without it, the shield may collect noise instead of blocking it.

4.3.5 COAXIAL CABLE

Coaxial cable uses a central conductor that shares the same axis.

It consists of:

- 1 **Copper Conductor** – carries the signal.
- 2 **Insulation** – layer of flexible plastic around the conductor.
- 3 **Shield** – woven copper braid or metal foil around the insulation. Reduces outside electromagnetic interference.
- 4 **Outer Jacket** – protects the cable from physical damage.



COMMON APPLICATIONS OF COAXIAL CABLE



Wireless Installations

(antennas to wireless devices)
Carries radio frequency (RF) energy between antennas and base equipment.



Cable Internet Installations

Provides internet connectivity by linking customer premises to the service provider using coaxial cable.

COMMON COAXIAL CONNECTORS



BNC



N type



F type



NOTE

Although UTP replaced coaxial in most Ethernet networks, coaxial is still used in wireless and cable internet systems.

4.4 UTP CABLING

4.4.1 PROPERTIES OF UTP CABLING



- UTP (Unshielded Twisted-Pair) is the standard for use in LANs.
- Consists of four pairs of color-coded copper wires that are twisted together and encased in a flexible plastic sheath.
- Small size and easy to install.
- Does not use shielding to block EMI/RFI. Instead, it relies on the cancellation effect of twisted pairs.

CANCELING UNWANTED INTERFERENCE



Cancellation

Two wires in a pair carry equal and opposite magnetic fields, so they cancel each other, reducing external EMI and RFI.



Varying the Number of Twists

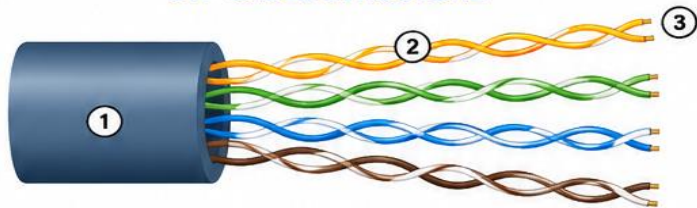
The number of twists in each pair is varied (as per standard) to further improve cancellation and reduce crosstalk.



KEY POINT

UTP cable relies on the cancellation effect produced by twisted wire pairs to limit signal degradation and provide self-shielding.

UTP CABLE STRUCTURE



1 **Outer Jacket** – protects the copper wires from physical damage.

2 **Twisted Pairs** – reduce interference (EMI and crosstalk) from other wires.

3 **Color-coded Plastic Insulation** – electrically isolates wires from each other and identifies each pair.

T568B COLOR CODE

Pair 1	White/Orange – Orange
Pair 2	White/Green – Green
Pair 3	White/Blue – Blue
Pair 4	White/Brown – Brown

RJ-45 CONNECTOR (PLUG)



- RJ-45 is the male connector crimped at the end of UTP cable.
- TIA/EIA-568 standard defines wire color codes for pin assignments (pinouts) of Ethernet cables.

4.4.2 UTP CABLING STANDARDS AND CONNECTORS

UTP cabling follows standards established by TIA/EIA. The key standard for LANs is TIA/EIA-568.

TIA/EIA-568 defines:

- ✓ Cable types
- ✓ Cable lengths
- ✓ Connectors
- ✓ Cable termination
- ✓ Methods of testing cable

IEEE rates UTP cables by performance categories based on bandwidth capacity:

- **Category 3** – Originally for voice; later used for data.
- **Category 5 and 5e** – Support 100 Mbps and 1000 Mbps.
- **Category 6** – Supports up to 10 Gbps.
- **Category 7** – Also support up to 10 Gbps.
- **Category 8** – Supports 40 Gbps.



Some manufacturers make cables exceeding Category 6a specifications and refer to these as Category 7.

POORLY TERMINATED UTP CABLE



- ✗ Wires are exposed and untwisted.
- ✗ Sheath does not cover the wires.
- ✗ Can cause crosstalk, signal loss and performance degradation.

PROPERLY TERMINATED UTP CABLE



- ✓ Wires are untwisted only as much as needed.
- ✓ Sheath goes up to the connector.
- ✓ Ensures better performance and reliability.

RJ-45 SOCKET (FEMALE)

Used on network devices, wall plates, patch panels, and other outlets.



4.4.3 STRAIGHT-THROUGH AND Crossover UTP CABLES

STRAIGHT-THROUGH CABLE

Most common cable. Used to connect unlike devices.

Examples:

Host ↔ Switch, Switch ↔ Router

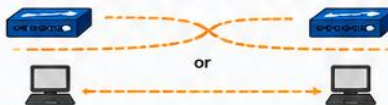


Crossover CABLE

Used to connect like devices.

Examples:

Switch ↔ Switch, Host ↔ Host, Router ↔ Router



NOTE

Modern devices (most NICs, switches, routers) support Auto-MDIX, which automatically detects the cable type and makes the internal connection.

T568A AND T568B STANDARDS



Both standards use the same colors but in different pin orders. T568B is more commonly used today.

CABLE TYPES AND STANDARDS

Cable Type	Standard	Application
Ethernet Straight-through	Both ends T568A or both ends T568B	Connects a network host to a network device such as a switch or hub
Ethernet Crossover	One end T568A, other end T568B	Connects two network hosts Connects two network intermediary devices (switch to switch or router to router)
Rollover	Cisco proprietary	Connects a workstation serial port to a router console port, using an adapter



4.5 FIBER OPTIC CABLING (PART 1)

4.5.1 PROPERTIES OF FIBER-OPTIC CABLING



- Fiber-optic cabling is the standard for use in networks.
- It is not as common as copper cabling but offers significant advantages.
- Fiber optic is immune to EMI and RFI (electromagnetic interference and radio frequency interference).
- Carries streams of light instead of electrical signals.
- Can support higher data rates and longer distances.
- Bandwidth is much greater and signal loss (attenuation) is lower than copper.
- Suitable for long-haul and high-speed networking.



KEY POINT

Fiber optic transmits data as light, giving higher bandwidth, lower loss, and immunity to EMI/RFI.



Difference: Multimode fiber has more dispersion (spreading of light pulses), which increases signal loss over time. Therefore, **MMF** can only travel **up to 500 meters** before significant signal loss occurs.

4.5.2 TYPES OF FIBER MEDIA

Fiber-optic cables are broadly classified into two types:

- Single-mode fiber (SMF)
- Multimode fiber (MMF)

SINGLE-MODE FIBER (SMF)

Uses expensive laser technology to send a single ray of light.

Popular for long-distance links (spanning hundreds of kilometers) such as long-haul telephone and cable TV applications.

Produces single, straight path for light.



Glass Core diameter = 9 μm

Glass Cladding diameter = 125 μm

Polymeric Coating

MULTIMODE FIBER (MMF)

Uses LED emitters to send light pulses.

Light enters the fiber at different angles, allowing multiple paths.

Popular in LANs (Local Area Networks) because LEDs are low-cost.

Allows multiple paths for light.



Glass Core diameter = 50–62.5 μm

Glass Cladding diameter = 125 μm

Polymeric Coating

4.5.3 FIBER-OPTIC CABLING USAGE



Enterprise Networks

Provides fast, high-speed backbone connections between switches, routers, and data centers.



Long-Distance Telecommunication

Ideal for long-haul telephony and cable TV networks where low signal loss and high bandwidth are required.



Data Centers & High-Speed Links

Used in data centers and between buildings for high throughput and reliability.



Immunity to Interference

Operates reliably in environments with high EMI/RFI where copper cabling may fail.

Fiber-optic cabling is the best choice where high speed, long distance, and reliability are required.

4.5.4 SUGGESTED USAGE GUIDELINES



Distance

Use SMF for very long distances (hundreds of kilometers).
Use MMF for shorter distances (up to 500 meters).



Cost

SMF is more expensive (laser based).
MMF is cost-effective (LED based).



Bandwidth

Both support high bandwidth, but SMF provides higher capacity over longer segments.



Environment

Fiber optic is immune to EMI and RFI, making it reliable in electrically noisy environments.

QUICK COMPARISON – SMF vs MMF

Feature	Single-Mode Fiber (SMF)	Multimode Fiber (MMF)
Light Source	Laser	LED
Mode of Light	Single straight path	Multiple paths (different angles)
Core Diameter	9 μm	50 – 62.5 μm
Typical Distance	Hundreds of kilometers	Up to 500 meters
Cost	Higher	Lower
Common Use	Long-haul telephony, cable TV	LANs, data centers



KEY TAKEAWAY

- ✓ Fiber optic uses light, not electricity.
- ✓ SMF = long distance, high capacity.
- ✓ MMF = shorter distance, cost-effective.
- ✓ Immune to EMI/RFI = more reliable.
- ✓ Choose fiber for speed, distance and performance!



4.5 FIBER OPTIC CABLING (PART 2)

4.5.4 FIBER-OPTIC CONNECTORS



An optical-fiber connector terminates the end of an optical fiber. Many types are available.



Connectors differ in size (dimensions) and coupling method.



Business choice depends on equipment.



NOTE:

Some switches and routers use small form-factor pluggable (SFP) transceivers to support fiber-optic connections. Search the Internet for various types of SFPs.

COMMON FIBER-OPTIC CONNECTOR TYPES

Straight-Tip (ST) Connectors	Subscriber Connector (SC) Connectors	Lucent Connector (LC) Simplex Connectors	Duplex Multimode LC Connectors



KEY FACTS

- ST connectors were among the first used. They lock with a twist-on/twist-off bayonet-style mechanism.
- For duplex operation, two fibers must be used (one for transmit, one for receive).
- Some fiber cables use a single connector (simplex); others use a duplex connector (e.g., Duplex Multimode LC).

4.5.5 FIBER PATCH CORDS



Fiber patch cords are used to interconnect infrastructure devices.

Color coding distinguishes types:

- Yellow – Single-mode fiber
- Orange – Multimode fiber



Fiber cables should be protected with a small plastic cap when not in use.

COMMON FIBER PATCH CORD TYPES

SC-SC Multimode Patch Cord	LC-LC Single-Mode Patch Cord	ST-LC Multimode Patch Cord	SC-ST Single-Mode Patch Cord



IMPORTANT: Fiber patch cords use a pair of connectors on each end to connect devices (e.g., switch to switch, switch to router).

4.5.6 FIBER VERSUS COPPER

Fiber-optic cable offers many advantages over copper cable.



Higher bandwidth

Fiber supports much higher speeds.



Much longer distance

Fiber can travel from 1 to 100,000 meters without signal loss.



Immune to EMI and RFI

Fiber is completely immune.



Immune to electrical hazards

Fiber does not conduct electricity.



Higher cost

Fiber media and connectors cost more.



Higher installation requirements

Installation skills required are higher.



Higher safety precautions

More care is needed when handling fiber.

IMPLEMENTATION ISSUES	UTP CABLING	FIBER-OPTIC CABLING
Bandwidth supported	10 Mb/s – 10 Gb/s	10 Mb/s – 100 Gb/s
Distance	Relatively short (1 – 100 meters)	Relatively long (1 – 100,000 meters)
Immunity to EMI and RFI	Low	High (completely immune)
Immunity to electrical hazards	Low	High (completely immune)
Media and connector costs	Lowest	Highest
Installation skills required	Lowest	Highest
Safety precautions	Lowest	Highest



KEY TAKEAWAY

Use fiber-optic cabling when you need high bandwidth, long distance, and immunity from EMI/RFI, such as in enterprise backbones, data centers, and harsh environments. Use UTP when cost is critical and distances are short.



4.6 WIRELESS MEDIA



4.6.1 PROPERTIES OF WIRELESS MEDIA



Wireless media is the **third way** to connect to the **physical layer** of a network (you may be using a tablet or smartphone).



Carries electromagnetic signals using **radio** or microwave frequencies to represent binary data.



Provides the **greatest mobility** and the **number of wireless-enabled devices** continues to increase.

LIMITATIONS OF WIRELESS



Coverage area – Works well in open areas. Buildings, construction materials, and local terrain can limit coverage.



Interference – Susceptible to interference from devices such as cordless phones, fluorescent lights, microwave ovens, and other wireless systems.



Security – Media is open; not authorized users can attempt access to the network. Security is a major part of wireless network administration.



Shared medium – WLANs operate in half-duplex (one device can send or receive at a time). Many users sharing the WLAN reduces bandwidth per user.

HOW WIRELESS COMMUNICATION WORKS



KEY POINT

Although wireless is increasing in popularity for desktop connectivity, copper and fiber are still the most common physical layer media for intermediate devices such as routers and switches.

4.6.2 TYPES OF WIRELESS MEDIA

IEEE and telecommunications standards for wireless data communications cover both the data link and physical layers.

PHYSICAL LAYER SPECIFICATIONS INCLUDE:



Data to radio signal encoding



Frequency and power of transmission



Signal reception and decoding requirements



Antenna design and construction

WIRELESS STANDARDS



Wi-Fi (IEEE 802.11)

- Wireless LAN (WLAN) technology, commonly called Wi-Fi.
- Uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).
- The wireless NIC listens first. If channel is clear, it transmits; otherwise it waits.
- Trademark of the Wi-Fi Alliance; used with IEEE 802.11 certified WLAN devices.



Bluetooth (IEEE 802.15)

- Wireless Personal Area Network (WPAN), commonly called Bluetooth.
- Uses device pairing to communicate over short distances (1 to 100 meters).



WiMAX (IEEE 802.16)

- Worldwide Interoperability for Microwave Access (WiMAX).
- Uses point-to-multipoint topology to provide wireless broadband access.



Zigbee (IEEE 802.15.4)

- For low-data rate, low-power communications.
- Designed for short-range, low data-rates and long battery life.
- Used in industrial and IoT environments (e.g., wireless light switches, medical device data collection).



NOTE: Other wireless technologies (e.g., cellular, satellite) can also provide data network connectivity, but they are out of scope for this module.

4.6.3 WIRELESS LAN

A common wireless data implementation is enabling devices to connect wirelessly via a LAN (WLAN). A WLAN requires these network devices:



Wireless Access Point (AP)

Concentrates the wireless signals from users and connects to the existing copper-based network infrastructure (e.g., Ethernet). Integrates functions of router, switch, and access point into one device.



Wireless NIC adapters

Provide wireless communication capability to network hosts (clients).

TYPICAL WLAN SETUP



BENEFITS OF WIRELESS DATA COMMUNICATION

- ✓ Cost savings on cabling and hardware
- ✓ Easy to deploy and expand
- ✓ Convenience and mobility
- ✓ Supports various devices



KEY TAKEAWAY

Wireless media provides mobility and convenience but has limitations in coverage, interference, security, and shared bandwidth. Choose the right wireless technology (Wi-Fi, Bluetooth, WiMAX, Zigbee) based on distance, data rate, power needs, and application requirements.